

WORK EXPERIENCE

- Chief Executive Officer, Stamp (YC W25)

Jan 2025 - Present

  - CEO and Founder of Stamp, the AI Secretary for email and calendar
  - Architected and Implemented a **service oriented backend** integrating directly with IMAP and SMTP email protocols, supporting all modern email client features (schedule send, snoozed emails, labels, etc)
  - Built an IMAP-optimized Agent SDK to construct a **multi-agent ecosystem**, with distinct agents for email annotation (summarization, todos, draft replies), user memories, Deep Research, Voice Mode, and more
  - Fine-tuned a multilingual embedding model via contrastive learning to implement both a **Retrieval Augmented Generation** pipeline (hybrid semantic search + reranking) and a **semi-supervised clustering** email labeling algorithm
  - Built **Web, iOS, Android, and Desktop apps**, with on-device persistence, notifications, and background refresh
  - Raised \$500k funding** and accepted into the **Y Combinator** Winter 2025 batch; managed all business operations in addition to engineering, including product ideation, marketing, hiring, etc.
- Software Engineer, Stripe

Jul 2024 - Dec 2024

  - Saved **\$1 million annually** in database costs by optimizing PaymentIntents API code, and performing shard-aware database migrations for **150 billion** stale records via PySpark, Databricks, Apache Airflow, and Apache Iceberg
  - Improved **API performance and stability** for the PaymentIntents Multicapture API by using the Factory Design Pattern to decouple code across customer environments, increasing test coverage by 20%
  - Improved Adaptive Pricing API semantics by adding presentment currency fields to the PaymentIntent Dataview and refactoring legacy PaymentIntent APIs to support multi-currency presentment
- Software Engineer Intern, Apple

May 2023 - Aug 2023

  - Architected 10,000+ line C++ library from scratch to detect topological and geometric changes in 3D Maps for Autonomous Vehicles, **reducing manual map annotation time by 70%**
  - Constructed custom UI geometry classes, modals, and event handlers in Objective C to visualize library analysis in the data labeling MacOS application
  - Implemented Protobuf schemas for library analysis serialization, visualization, and deployment

RESEARCH

- Computer Vision Researcher, Cornell University

Aug 2023 - May 2024

  - Worked with Prof. Noah Snavely on two projects centered around Multi-View Reconstruction
  - Implemented **CoTracker optical flow pipeline** to preprocess video data for **generative AI video editing**
  - Investigated SoTA **dust3r** model architectural variants capable of EXIF metadata processing during reconstruction

PROJECTS

- Archivist Code Search:

Created a VS Code Extension to provide blazingly fast Semantic Search for code. Used VS Code APIs for realtime indexing, deployed a fine-tuned embedding model, and implemented syntax tree aware chunking.

Tech: FastAPI, HuggingFace SentenceTransformers, MongoDB Atlas Vector Search, AWS EC2, Docker
- Small Language Model (SLM):

Architected and trained a **1B parameter dense transformer from scratch**, performing distributed pretraining on 5B tokens and Supervised Fine Tuning on chat data and tool calling. Implemented positional embeddings via RoPE and YaRN, KV Cache for inference, and built a Mixture of Experts (MoE) variant.

Tech: PyTorch, HuggingFace Tokenizers, HuggingFace Datasets, AWS EC2, WandB
- Strompt (structured prompts):

A lightweight extensible Prompt Rendering Library, published on NPM. Enables readable, maintainable, typo-free prompt generation for agents via the Fluid Builder Design Pattern.

Tech: TypeScript, GitHub Actions, NPM

EDUCATION

- Cornell University

Ithaca, NY

B.S. Computer Science; GPA: 4.16/4.0

2021 - 2024

Minor: Applied Mathematics

Awards: Merrill Presidential Scholar (Awarded for achievement & leadership to top 8 CS Undergraduates)

Courses: Honors Object Oriented Data Structures, Large Scale Machine Learning, Analysis of Algorithms, Computer Vision

Activities: Cornell AppDev (President), Cornell Hack4Impact (Engineering Lead), Association of CS Undergrads (Corporate Officer)

SKILLS

|                  |                                                                                  |
|------------------|----------------------------------------------------------------------------------|
| Languages        | Java, TypeScript, Python, C++, HTML/CSS/JavaScript, C, OCaml, LaTeX              |
| Machine Learning | PyTorch, SkLearn, SentenceTransformers, NLTK, Pandas, Numpy, Jupyter             |
| Frameworks       | Express & Node.js, FastAPI, Django, Flask, React, React Native, Next.js          |
| AI Tooling       | Model Context Protocol (MCP), Agent Skills, Cursor, Claude Code, Codex, OpenClaw |
| DevOps & DB      | MongoDB, PostgreSQL, Docker, Azure, AWS, GCP, Protobuf, Jenkins, Terraform       |